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Matter RGB Light with the Arduino Nano Matter

Learn how to build a Matter RGB light using the Arduino Nano Matter

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Overview

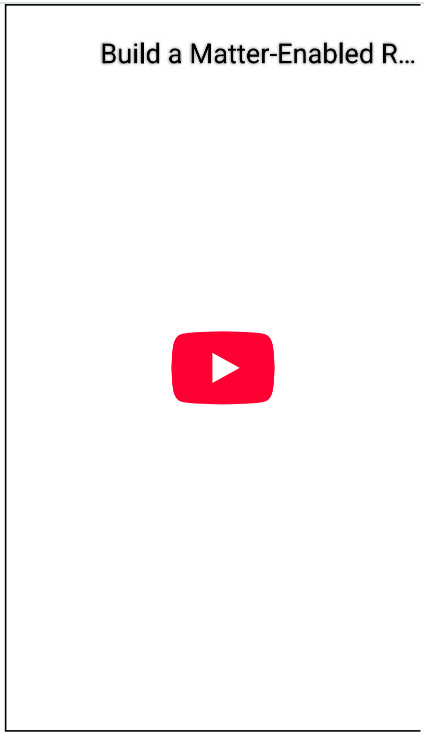
This tutorial will teach you how to us the Arduino Nano Matter to create a Matter RGB Lightbulb to light up any room with colors.

RGB light overview

Thanks to the seamless compatibility of the Nano Matter with almost any Matter network we can easily integrate our RGB light with Amazon Alexa, Google Assistant, Apple Home Home Assistant and even any professional custom solution.

We have prepared a short demo in video format in case you are a visual learner.

Help



Hardware and Software Requirements

Hardware Requirements

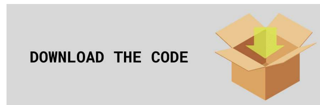
- Arduino Nano Matter (x1)
- Grove - 8x8 RGB LED Matrix (x1)
- Breadboard (x1)
- Jumper wires
- Eero 6+ WiFi Extender (Thread Border Router) (x1)
- USB-C® cable (x1)

Software Requirements

- Arduino IDE 2.0+ or Arduino Cloud Editor
- Amazon Alexa
- Seeed_RGB_LED_Matrix library

You can install it as .ZIP using the Arduino IDE.

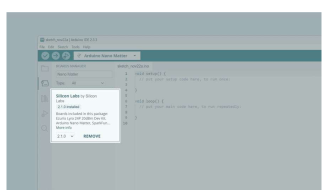
Download the Project Code



Download the complete project code here.

Board Core and Libraries

The **Silicon Labs** core contains the libraries and examples you need to work with the board's components, such as its Matter, Bluetooth® Low Energy, and I/Os. To install the Nano Matter core, navigate to **Tools > Board > Boards Manager** or click th Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for `Nano Matter` and install the latest `Silicon Labs` core version.

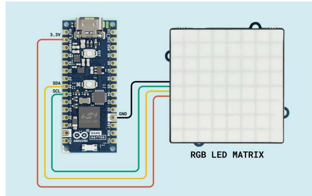


Installing the Silicon Labs core in the Arduino IDE

Project Setup

Schematic Diagram

Use the following connection diagra for the project:

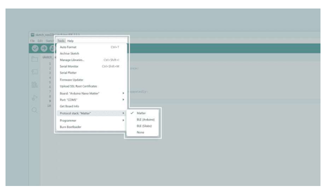


Project wiring diagram

The RGB LED matrix is powered by the Nano Matter `3.3V` output pin, and it is controlled using an I2C connection.

Programming

In the Arduino IDE upper menu, navigate to **Tools > Protocol stack** and select **Matter**.



Matter Protocol stack selected

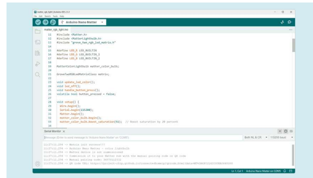
Copy and paste the following sketch

```
#include <Matter.h>
#include <MatterLightbu
#include "grove_two_rgl

#define LED_R LED_BUIL
#define LED_G LED_BUIL
#define LED_B LED_BUIL

MatterColorLightbulb ma

GroveTwoRGBLedMatrixCla

void update_led_color()
void led_off();
void handle_button_pres
volatile bool button_pr

void setup() {
  Wire.begin();
  Serial.begin(115200);
  Matter.begin();
  matter_color_bulb.beg
  matter_color_bulb.boo

  // Set up the onboard
  pinMode(BTN_BUILTIN,
  attachInterrupt(BTN_E

```

The structure of this example code is very simple, the main functions are explained below:

- In the `setup()` function we initialize the Matter connectivity and the RGB Matrix I2C communication.
- In the `loop()` function we listen to any request of controlling the lightbulb state or color and update the matrix respectively.
- There are some other functions to handle button press or RGB matrix color setup.

Once you uploaded the example code to the Nano Matter, open the Serial Monitor and reset the board.



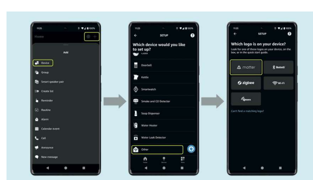
QR Code URL

After the reset you will find on the serial port the URL that generates th QR for the Matter device commissioning.

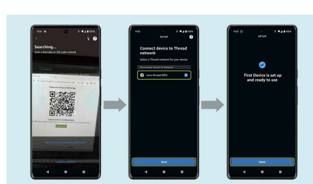
Adding the Device (Commissioning)

Copy and paste the QR code URL on your favorite web browser and a unique QR code will be generated fo your board.

Go to your **Amazon Alexa** app, navigate to **Devices > Add Device > Other**, select the **Matter** option anc scan the QR code.



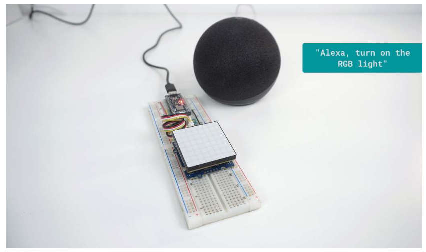
Adding the device to Google Home app



Your device is successfully added

Final Results

Finally, you will be able to control th
RGB light from your smartphone, hu
or asking your personal assistant.



Conclusion

In this tutorial we have learned how to create a Matter enabled RGB light that can be controlled from our smartphone and personal assistant. The Nano Matter allows us to seamlessly integrate our own light a a commercial product with our current smart home ecosystem.

Next Steps

You can take this solution even further by adding fading animations or even using the matrix to display text or graphics.

Was this article helpful?